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CUTTING TOOL HAVING A CUTTING HEAD AND AN ADAPTER HAVING A CLAMPING MECHANISM THEREOF

Abstract

A cutting tool having an adapter and a cutting head. The adapter has an adapter clamping portion and a clamping mechanism. The clamping mechanism has an actuator, a central wedge and two flanges slidingly engaged to the central wedge. Each of the two flanges has an elongated rear slanted flange portion configured for clamping the cutting head.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims benefit of, and priority to, U.S. Provisional Patent Application Ser. No. 63/559,414, filed Feb. 29, 2024, the entire contents of which are hereby incorporated by reference.

FIELD OF THE INVENTION

[0002] The subject matter of the present application relates to clamping mechanisms, in general, and in particular to cutting heads releasably secured to an adapter using a clamping mechanism.

BACKGROUND OF THE INVENTION

[0003] In the field of metal cutting, interchangeable cutting heads which are releasably secured to an adapter are known. For example, an interchangeable cutting head releasably secured to an adapter is shown in US2015174666.

[0004] Also known are force multiplier mechanisms used in clamping a first part to a second part. Force multiplier mechanisms refer to mechanisms where the clamping means (such as, for example, bearing balls or flanges) are located on opposing sides of an actuator which actuates (i.e. activates) the clamping means, enabling greater forces for clamping. For example, such force multiplier mechanisms can be seen in U.S. Pat. Nos. 4,736,659, 5,261,302, and 11,358,227.

SUMMARY OF THE INVENTION

[0005] The subject matter of this application may be an adapter with an improved force multiplier mechanism. Such an improved force multiplier mechanism may allow for more stable clamping.

[0006] In accordance with a first aspect of the subject matter of the present application there is provided an adapter having a central adapter axis Ca defining an outward radial direction Ro extending away from the central adapter axis Ca, opposing forward and rearward adapter directions Fa, Ra and a central plane Pc orthogonal to the central adapter axis Ca, the adapter **20** comprising: [0007] a rear adapter end section; [0008] a forward adapter end section located forward of the rear adapter end section and a peripheral adapter surface connecting the forward adapter end section and the rear adapter end section; [0009] the forward adapter end section including an adapter clamping portion, the adapter clamping portion comprising an adapter clamping receptacle, the adapter clamping receptacle defining: [0010] a first flange bore opening out to the peripheral adapter surface; [0011] a second flange bore opening out to the peripheral adapter surface; and [0012] an actuator bore extending into the adapter clamping receptacle;

and [0013] a clamping mechanism comprising: [0014] an actuator defining an actuating axis A having opposite first and second axial directions D**1**, D**2** extending parallel to the actuating axis A, the actuator being aligned with the actuator bore and being at least partially located in the adapter clamping receptacle; [0015] a central wedge connected to the actuator; [0016] a first flange slidingly connected to the central wedge and configured to project through the first flange bore; and [0017] a second flange opposite the first flange about the actuating axis A, the second flange slidingly connected to the central wedge and configured to project through the second flange bore; [0018] each of the first and second flanges comprising: [0019] a forward facing flange surface; and [0020] a rearward facing flange surface, the rearward facing flange surface comprising a rear slanted flange portion, the rear slanted flange portion extending forwardly in the outward radial direction Ro, the rear slanted flange portion elongated in a direction along the central plane Pc.

[0021] In accordance with a second aspect of the subject matter of the present application there is provided a cutting head defining a central head axis Ch defining forward and rearward head directions Fh, Rh, the cutting head comprising: [0022] a forward head end section; [0023] a rearward head end section located rearward of the forward head end section and a peripheral head surface connecting the forward head end section and the rearward head end section; [0024] the rearward head end section including a head clamping portion, the head clamping portion comprising a receptacle wall, the receptacle wall including a forward facing head clamping surface and a rearward facing head abutment surface, the head clamping surface extending in a direction away from the central head axis Ch and in the forward head direction Fh, the receptacle wall defining: [0025] a head clamping receptacle recessed into the head clamping portion; and [0026] an actuator access bore opening out to the peripheral head surface and to the head clamping receptacle; [0027] the peripheral head surface including an integral cutting element or defining a cutting insert pocket (**152**) configured to receive a cutting element.

[0028] In accordance with a third aspect of the subject matter of the present application there is provided a cutting tool comprising [0029] an adapter; and [0030] the cutting head according to the second aspect.

[0031] In accordance with a fourth aspect of the subject matter of the present application there is provided a cutting tool comprising [0032] an adapter according to the first aspect; and [0033] a cutting head attached to the adapter, the cutting head defining a central head axis coinciding with the central adapter axis of the adapter and having forward and rearward head directions extending parallel to the central head axis.

[0034] It is understood that the above-said is a summary, and that features described hereinafter may be applicable in any combination to the subject matter of the present application, for example, any of the following features may be applicable to the cutting tool and/or the adapter and/or the cutting head. The rear slanted flange portion may be cone shaped. Each of the first and second flanges may comprise: a wedge connector portion located adjacent to the central wedge, the wedge connector portion connecting the forward and rearward facing flange surfaces, the wedge connector portion of the first flange slants towards the actuating axis A in the first axial direction D**1** and the wedge connector portion of the second flange slants towards the actuating axis A in the first axial direction D**1**. A wedge connector angle wca, defined between the actuating axis A and the wedge connector portion of one of the first and second flanges, may fulfil the following condition: 5°≤wca≤25°. Each of the first and second flanges may comprise, at the wedge connector portion, a flange dovetail formation, the central wedge comprises two wedge dovetail formations engaging with the flange dovetail formations; each of the flange dovetail formations in sliding engagement with one of the two wedge dovetail formations in a direction along the actuating axis A. The actuator may have an active position in which the first and second flanges project radially from the first and second flange bores, respectively, wherein the actuator has an inactive position in which the first and second flanges do not project radially from the first and second flange bores, respectively. Each of the first and second flanges may further comprise: an outer flange surface facing away from the actuating axis A, wherein in the inactive position, the outer flange surface of each of the first and second flanges is level with the peripheral adapter surface adjacent to the respective one of the first and second flange bores. A rear slanted flange angle rsa, defined between the central plane Pc and the rear slanted flange portion, may fulfil the following condition: 3°≤rsa≤20°. The clamping mechanism may further comprise an actuator captive mechanism securing the actuator in the adapter. The rearward facing flange surface may be planar. The first and second flanges may be identical. The central wedge may comprises: a first wedge surface; a second wedge surface located farther in the second axial direction D**2** than the first wedge surface; a forward wedge surface connecting the first and second wedge surfaces; a rearward wedge surface located rearwardly of the forward wedge surface and connecting the first and second wedge surfaces; an actuator wedge section extending between the first and second wedge surfaces and engaged to the actuator; a first flange connector portion slidingly connected to the first flange, the first flange connector portion extending between the first and second wedge surfaces; and a second flange connector portion slidingly connected to the second flange, the second flange connector portion extending between the first and second wedge surfaces. The first flange connector portion may slant towards the actuating axis A in the first axial direction D**1**; and the second flange connector portion may slant towards the actuating axis A in the first axial direction D**1**. A flange connector angle fca, defined between the first and second flange connector portions, may fulfil the following condition: 10°≤fca≤50°. The wedge and flange connector angles wca, fca may fulfil the following condition: fca=2*wca. Each of the first and second flanges may comprise a wedge dovetail formation. The head clamping surface may be elongated in a direction about the central head axis Ch. The head clamping surface may encircle the central head axis Ch. The head clamping surface may be cone shaped. A head clamping angle hca, defined between the head clamping surface and a head plane Ph, the head plane Ph being perpendicular to the central head axis Ch, may fulfil the following condition: 5°≤hca≤22°. The head abutment surface may comprise: two rear abutting portions and two relief portions located forwardly of the two rear abutting portions, each of the two relief portions being located between the two rear abutting portions.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0035] For a better understanding of the present application and to show how the same may be carried out in practice, reference will now be made to the accompanying drawings, in which:

[0036] FIG. **1** is a perspective view of a cutting tool according to an aspect of the present application including an adapter and a cutting head;

[0037] FIG. **2** is an exploded view, with parts separated, of the cutting tool shown in FIG. **1**;

[0038] FIG. **3** is a perspective view of the adapter shown in FIG. **1**;

[0039] FIG. **4** is a front view of the adapter shown in FIG. **3**;

[0040] FIG. **5** is a rear view of the adapter shown in FIG. **3**;

[0041] FIG. **6** is a flange-side view of the adapter shown in FIG. **3**;

[0042] FIG. **7** is a cross-sectional view taken along line A-A of the adapter shown in FIG. **6**;

[0043] FIG. **8** is a cross-sectional view taken along line B-B of the adapter shown in FIG. **6**;

[0044] FIG. **9** is an actuator side view of the adapter shown in FIG. **3**;

[0045] FIG. **10** is a cross-sectional view taken along line C-C of the adapter shown in FIG. **9**;

[0046] FIG. **11** is a perspective view of a clamping mechanism shown in FIG. **2**;

[0047] FIG. **12** is an exploded view, with parts separated, of the clamping mechanism shown in FIG. **11**;

[0048] FIG. **13***a* is a perspective view of a central wedge shown in FIG. **11**;

[0049] FIG. **13***b* is a first side view of the central wedge shown in FIG. **13***a;*

[0050] FIG. **13***c* is an axial view of the central wedge shown in FIG. **13***a;*

[0051] FIG. **13***d* is a second side view of the central wedge shown in FIG. **13***a;*

[0052] FIG. **13***e* is a cross-sectional view taken along line D-D of the central wedge shown in FIG. **13***d;*

[0053] FIG. **14***a* is a perspective view of one of a first and second flanges shown in FIG. **11**;

[0054] FIG. **14***b* is an axial view of the flange shown in FIG. **14***a;*

[0055] FIG. **14***c* is an outer radial view of the flange shown in FIG. **14***a;*

[0056] FIG. **14***d* is a wedge connector view of the flange shown in FIG. **14***a;*

[0057] FIG. **14***e* is a first side view of the flange shown in FIG. **14***a;*

[0058] FIG. **14***f* is a second side view of the flange shown in FIG. **14***a;*

[0059] FIG. **15** is an actuator access view of a cutting head without insert pockets yet formed thereon and according to an aspect of the present invention;

[0060] FIG. **16** is a cross-sectional view taken along line E-E of the cutting head shown in FIG. **15**;

[0061] FIG. **17** is a rear view of the cutting head shown in FIG. **15**;

[0062] FIG. **18** is an actuator side view of the cutting tool shown in FIG. **1**;

[0063] FIG. **19** is a flange side view of the cutting tool shown in FIG. **1**;

[0064] FIG. **20** is an active position cross-sectional view taken along line F-F of the cutting tool

shown in FIG. **19**;
[0065] FIG. **21** is an inactive position cross-sectional view taken along line F-F of the cutting tool shown in FIG. **19**;
[0066] FIG. **22** is a perspective view of a cutting tool without insert pockets yet formed on the cutting head, according to an aspect of the present invention; and
[0067] FIG. **23** is an exploded view, with parts separated of the cutting tool shown in FIG. **22**.
[0068] It will be appreciated that for simplicity and clarity of illustration, elements shown in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity, or several physical components may be included in one functional block or element. Further, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements.
DETAILED DESCRIPTION OF THE INVENTION
[0069] In the following description, various aspects of the subject matter of the present application will be described. For purposes of explanation, specific configurations and details are set forth in sufficient detail to provide a thorough understanding of the subject matter of the present application. However, it will also be apparent to one skilled in the art that the subject matter of the present application can be practiced without the specific configurations and details presented herein.
[0070] Attention is first drawn to FIGS. **1** and **2**, showing a cutting tool **1** having a cutting head **120** releasably attached to an adapter **20** via a clamping mechanism **80**. The clamping mechanism **80** may synonymously be referred to as a clamping insert, a clamping arrangement, or an expandable wedge herein. The cutting head **120** can accommodate cutting elements (not shown). For example, such cutting elements may be integrally formed cutting teeth or, more typically, interchangeable cutting inserts. Such cutting elements may be made from hard metals, such as cemented carbide. The cutting elements may be made of harder material than the cutting head **120**.
[0071] Drawing attention to FIGS. **3** to **10**, the adapter **20** has a central adapter axis Ca. The central adapter axis Ca defines an outward radial direction Ro, a forward adapter direction Fa and a rearward adapter direction Ra opposite to the forward adapter direction Fa. The outward radial direction Ro is perpendicular to the central adapter axis Ca, from the central adapter axis Ca outwards. The forward and rearward adapter directions Fa, Ra may be parallel to the central adapter axis Ca.
[0072] The adapter **20** includes a rear adapter end section **24**, a forward adapter end section **30**, a peripheral adapter surface **70** and a clamping mechanism **80**. The forward adapter end section **30** is located forward of the rear adapter end section **24**. The peripheral adapter surface **70** connects the rear and forward adapter end sections **24**, **30**.
[0073] The forward adapter end section **30** includes an adapter clamping portion **34**. The adapter clamping portion **34** includes an adapter clamping receptacle **38** and defines a first flange bore **42**, a second flange bore **44**, and an actuator bore **66**. As best seen in FIG. **10**, the adapter clamping receptacle **38** is at least partially hollow. The adapter clamping receptacle **38** defines a hollow portion **40**. The hollow portion **40** may allow at least part of the clamping mechanism **80** to be placed therein. In some embodiments, the adapter clamping receptacle **38** is centered about the central adapter axis Ca. In some embodiments, the adapter clamping portion **34** further includes an adapter torque transfer mechanism **58**. Such an adapter torque transfer mechanism **58** may be advantageous, e.g., for rotary cutting tools.
[0074] The adapter clamping portion **34** further includes an adapter abutment surface **60**. The adapter abutment surface **60** faces in the forward adapter direction Fa. The adapter abutment surface **60** encircles (i.e. surrounds) the central adapter axis Ca. In some embodiments, the adapter abutment surface **60** encircles the adapter clamping receptacle **38**. In such embodiments, the adapter abutment surface **60** may delimit the adapter clamping portion **34** in the rearward adapter

direction Ra. Having the adapter abutment surface **60** encircle the adapter clamping receptacle **38** may allow for a more stable clamping between the adapter **20** and the cutting head **120**.

[0075] Each of the first and second flange bores **42**, **44** are formed in the adapter clamping receptacle **38**. Each of the first and second flange bores **42**, **44** open out to the peripheral adapter surface **70**. Further, each of the first and second flange bores **42**, **44** open out to the hollow portion **40** of the adapter clamping receptacle **38**. In some embodiments the first and second flange bores **42**, **44** are located on opposite sides of the central adapter axis Ca. Specifically, the first and second flange bores **42**, **44** are opposite one another relative to the central adapter axis Ca.

[0076] In some embodiments, each of the first and second flange bores **42**, **44** is at least partially defined by a forward bore surface **48** and/or a rearward bore surface **54** (see FIG. **6**). The forward bore surface **48** faces in the rearward adapter direction Ra. The rearward bore surface **54** is located rearwardly of, and faces in the direction of, the forward bore surface **48**. Each of the forward and rearward bore surfaces **48**, **54** may be orthogonal to the central adapter axis A. The forward and rearward bore surfaces **48**, **54** may oppose one another.

[0077] In some embodiments, the forward bore surface **48** includes a chamfered bore portion **50**. The chamfered bore portion **50** is slanted in the forward adapter direction Fa and in the outward radial direction Ro. This chamfered bore portion **50** may ensure that the clamping mechanism **80** (and specifically a first flange **100** and a second flange **102** as discussed below) does not deform the forward bore surface **48** when securing the cutting head **120** to the adapter **20**.

[0078] The actuator bore **66** extends into the adapter clamping receptacle **38**. Specifically, the actuator bore **66** opens out to at least one of the hollow portion **40** or the peripheral adapter surface **70**. The actuator bore **66** may open out to both the hollow portion **40** and the peripheral adapter surface **70**, allowing an actuator **82**, as will be discussed below, to pass therethrough. In some embodiments, the first and second flange bores **42**, **44** are located on opposite sides of the central adapter axis Ca and on opposite sides of the actuator bore **66**.

[0079] As best seen in FIGS. **11** to **14***f*, the clamping mechanism **80** includes an actuator **82**, a central wedge **90**, a first flange **100**, and a second flange **102**. The actuator **82** defines an actuating axis A. The actuating axis A extends in a first axial direction D**1** and an opposite second axial direction D**2**. The actuator **82** is aligned with the actuator bore **66** and is at least partially located in the adapter clamping receptacle **38**. Specifically, the actuator bore **66** may open out to the hollow portion **40** of the adapter clamping receptacle **38**.

[0080] A key (not shown) actuates the actuator **82**. The actuator **82** may be rotated about the actuating axis A in a first or fastening direction Df and an opposite second or releasing direction Dr (see FIG. **9**). In some embodiments the actuator **82** may be, as seen in the attached figures, a screw. In some embodiments, the key causes the actuator **82** to rotate in the fastening and releasing directions Df, Dr. Alternatively, the key **100** may cause the actuator **82** to move in different directions, such as the first or second axial directions D**1**, D**2**, e.g., by exerting forces axially along the actuating axis A or by threads of the actuator **82** engaging surfaces disposed about the actuating axis A upon rotation of the actuator **82**.

[0081] In some embodiments, the actuating axis A is perpendicular to the central adapter axis Ca. Alternatively, the actuating axis A may be, for example, parallel to the central adapter axis Ca.

[0082] As shown in FIGS. **20** and **21**, the adapter **20** has an active position (seen in FIG. **20**) and an inactive position (seen in FIG. **21**). In the active position, the actuator **82** is actuated so that the first and second flanges **100**, **102** project radially or extend from, respectively, the first and second flange bores **42**, **44**. In the inactive position, the actuator **82** is actuated so that the first and second flanges **100**, **102** do not project radially or are retracted from, respectively, the first and second flange bores **42**, **44**. That is to say, in the active position the first and second flanges **100**, **102** extend farther in the outward radial direction Ro relative to the first and second flange bores **42**, **44**. Similarly, in the inactive position the first and second flange bores **42**, **44** are level with, or extend farther than, the first and second flanges **100**, **102** in the outward radial direction Ro. Having the first and second flanges **100**, **102** retracted into the adapter **20** when the actuator **82** is in the inactive position may allow, for example, for a thicker cutting head **120**, instead of, for example, creating grooves to accommodate partially protruding flanges therein. This may positively affect the stability of the cutting tool **1**.

[0083] In some embodiments, each of the first and second flanges **100**, **102** further include an outer flange surface **114**. The outer flange surface **114** faces away from the actuating axis A. The outer flange surface **114** may connect forward and rearward facing flange surfaces **106**, **110** as discussed below. In the inactive position (defined herein above as the inactive position of the adapter **20**), the outer flange surface **114** of each of the first and second flanges **100**, **102** is level with the peripheral adapter surface **70** adjacent to the respective one of the first and second flange bores **42**, **44**.

[0084] The first and second flanges **100**, **102** are located opposite to one another about the actuating axis A. That is to say, as seen in FIG. **8**, the first flange **100** is located on one side of the actuating axis A, and the second flange **102** is located on the other side of the actuating axis A. In some embodiments, the actuator **82**, the first flange **100** and the second flange **102** are level with one another along the central adapter axis Ca. In other embodiments, the first and second flanges **100**, **102** may be level with one another along the central adapter axis Ca with the actuator **82** being located either rearwardly or forwardly of the first and second flanges **100**, **102**.

[0085] The central wedge **90** is connected to the actuator **82**. Such a connection ensures that the central wedge **90** is moveable by the actuator **82**. The connection may be, for example, a threaded connection with the central wedge **90** and the actuator **82** each having threads therein. When the key actuates the actuator **82**, the central wedge **90** is moved. Specifically, the central wedge **90** may be moved axially along the actuating axis A. The first and second flanges **100**, **102** are in sliding connection with the central wedge **90**. That is to say, the first and second flanges **100**, **102** are secured to the central wedge **90** such that movement between each of the first and second flanges **100**, **102** and the central wedge **90** is allowed in some directions but is not allowed or restricted in other directions. As best shown in FIGS. **8**, **20**, and **21**, the first and second flanges **100**, **102** are moveable in a direction perpendicular to the actuating axis A, but are not moveable along the actuating axis A and the central adapter axis Ca.

[0086] The first and second flanges **100**, **102** are configured to project through the first and second flange bores **42**, **44**, respectively. That is to say, the first flange **100** is configured to project through the first flange bore **42** and the second flange **102** is configured to project through the second flange bore **44**. The first and second flanges **100**, **102** are at least partially located within the first and second flange bores **42**, **44**, respectively.

[0087] The above-described clamping mechanism **80** is a so-called “force multiplier”. Such a mechanism is capable of providing greater clamping forces than the force provided by the actuator **82** alone. This is due, in part, to the first and second flanges **100**, **102** moving in a direction transverse (i.e. not parallel) to the actuating axis A. There are additional factors facilitating such clamping forces, for example the angle of surfaces participating in the clamping of the adapter **20** and the cutting head **120**, as discussed below. The clamping mechanism **80** described herein may be desirable for rotary cutting tools (for example milling tools) which experience complex and varying cutting forces during machining operations. The adapter **20** and its clamping mechanism **80** may provide improved stability and great clamping forces than a clamping mechanism without first and second flanges and/or without the angle of surfaces as discussed below.

[0088] When the central wedge **90** is moved by the actuator **82**, the first and second flanges **100**, **102** slide along the central wedge **90** while still being located within the first and second flange bores **42**, **44**. The first and second flanges **100**, **102** are thus movable relative to the adapter **20**, but only in a direction different from the direction in which the central wedge **90** is moved. The movement of the first and second flanges **100**, **102** relative to the central wedge **90** may, for example, cause the first and second flanges **100**, **102** to either protrude out, or retract inwardly, of the peripheral adapter surface **70**.

[0089] In the active position, when the key actuates the actuator **82**, the central wedge **90** is moved, causing the first and second flanges **100**, **102** to move slidingly relative to the central wedge **90** and project outwards from their respective one of the first or second flange bores **42**, **44**. Likewise, in the inactive position, when the key actuates the actuator **82**, the central wedge **90** is moved, causing the first and second flanges **100**, **102** to move slidingly relative to the central wedge **90** and retract inwardly within their respective one of the first or second flange bores **42**, **44**.

[0090] In some embodiments, in the active position, the actuator **82** is actuated by rotation in the fastening direction Df. Then the central wedge **90** is moved in the first axial direction D**1**. Likewise, in the inactive position, the actuator **82** is rotated in the releasing direction Dr. Then the central wedge is moved in the second axial direction D**2** (see FIGS. **8** and **9**).

[0091] In some embodiments, where the actuator **82** is actuated through rotating about the actuating axis A, the clamping mechanism **80** may further include an actuator captive mechanism **84** (see FIGS. **11** and **12**). The actuator captive mechanism **84** secures the actuator **82** in the adapter **20**, ensuring that while the actuator **82** is movable in order to facilitate a clamping engagement between the adapter **20** and the cutting head **120**, the actuator **82** remains within the adapter **20**. In the embodiments shown in the Figures, the actuator **82** is a screw and the actuator captive mechanism **84** is a nut engaging the screw to ensure it does not detach from the adapter **20**. The actuator captive mechanism **84** is not confined to a nut or the like and may, for example, include a pin (not shown) engaging a recessed portion of a screw, the screw acting as the actuator **82**.

[0092] The following description will refer to features shared by both the first and second flanges **100**, **102** using the same numerals. In embodiments, the first and second flanges **100**, **102** correspond to the extent that they are interchangeable. The first and second flanges **100**, **102** may be identical to one another. It will be noted that in some embodiments, the first and second flanges **100**, **102** are both identical and interchangeable, as shown in FIGS. **14***a* to **14***f*. The flange shown embodies either of the first and second flanges **100**, **102**. In other envisioned embodiments the first and second flanges **100**, **102** are not required to be identical with respect to all features thereof.

[0093] Each of the first and second flanges **100**, **102** includes a forward facing flange surface **106** and a rearward facing flange surface **110**. The forward facing flange surface **106** faces in the forward adapter direction Fa. The rearward facing flange surface **110** faces in the rearward adapter direction Ra. The forward facing flange surface **106** may be opposite the rearward facing flange surface **110**. The rearward facing flange surface **110** includes a rear slanted flange portion **112**. The rear slanted flange portion **112** is a portion of the rearward facing flange surface **110** that extends forward in the outer radial direction Ro. That is to say, the rear slanted flange portion **112** faces in both the rearward adapter direction Ra and in the outward radial direction Ro. In some embodiments, the forward facing flange surface **106** includes an additional slanted flange surface **108** corresponding to the rear slanted flange surface **112**. This can facilitate using the first and second flanges **100**, **102** interchangeably. Further, in some embodiments, the forward and rearward facing flange surfaces **106**, **110** are identical to one another.

[0094] A central plane Pc is defined as orthogonal to the central adapter axis Ca. The rear slanted flange portion **112** is elongated in a direction along the central plane Pc. In some embodiments, the rear slanted flange portion **112** is specifically elongated in a direction circumferential to the adapter. Differently worded, the rear slanted flange portion **112** has a greater extent along a direction perpendicular to the central adapter axis Ca, relative to the extent of the rear slanted flange portion **112** in other directions. The rear slanted flange portion **112** extends about the central adapter axis Ca.

[0095] The rear slanted flange portion **112** being elongated in a direction along the central plane Pc allows two points of contact between the rear slanted flange portion **112** and the cutting head **120**. Having a two-point contact between the cutting head **120** and each of the first and second flanges **100**, **102** ensures increased stability in the clamping between the adapter **20** and the cutting head **120**. Additionally, the direction of elongation of the rear slanted flange portion **112** may further

ensures that the clamping forces arising from the abutment of the cutting head **120** and each of the first and second flanges **100**, **102** is directed mainly in the rearward adapter direction Ra which may increase the stability of the connection between the adapter **20** and the cutting head **120**.
[0096] In some embodiments, the rearward facing flange surface **110** is planar. Further, the rearward facing flange surface **110** may be parallel to the central plane Pc. When the rear slanted flange portion **112** clamps the cutting head **120**, the forward facing flange surface **106** and the rearward facing flange surface **110** abut against the forward and rearward bore surfaces **48**, **54**, respectively. Having the rearward facing flange surface **110** planar may ensure a more stable abutment between the first and second flanges **100**, **102** and the forward and rearward bore surfaces **48**, **54**. Note that the rearward facing flange surface **110** may be split, for example by a recessed portion. As explained above, such a planar surface may improve stability.
[0097] In some embodiments, the rear slanted flange portion **112** is cone shaped. As seen in the figures, the rear slanted flange portion **112** is slanted relative to the central adapter axis Ca, In embodiments where the rear slanted flange portion **112** is annular, the resulting geometric shape may be conical. Alternatively, the rear slanted flange portion **112** may be, for example, flat, chamfered, or recessed. The shape of the rear slanted flange portion **112** needs to conform to the shape of the relative feature of the cutting head **120** against which it abuts, which will be discussed below. A cone-shape may be desirable as it is a relatively simple shape to machine, and may result in high accuracy. This advantage may be relevant to both the first and second flanges **100**, **102** as well as the cutting head **120**. Additionally, such a shape may further assure of two-point contact between the rear slanted flange portion **112** and the cutting head **120**.
[0098] In some embodiments, a rear slanted flange angle rsa is defined between the central plane Pc and the rear slanted flange portion **112**. The rear slanted flange angle rsa may fulfil the following condition: $3° \leq rsa \leq 20°$. The rear slanted flange angle rsa may further fulfil the following condition: $5° \leq rsa \leq 15°$. Specifically, the rear slanted flange angle rsa may fulfil the following condition: $8° \leq rsa \leq 12°$.
[0099] The above-mentioned angle ranges for the rear slanted flange angle rsa may balance the direction of the clamping forces and the extent to which the first and second flanges **100**, **102** project. This is detailed as follows: [0100] For higher values of the rear slanted flange angle rsa, the clamping forces between the first and second flanges **100**, **102** and the cutting head **120** may be, for example, directed more along the outward radial direction Ro relative to lower values of the rear slanted flange angle rsa. Thus, clamping forces between the adapter and the cutting head **120** are lower for higher values of rear slanted flange angle rsa, relative to lower values of the rear slanted flange angle rsa. [0101] For lower values of the rear slanted flange angle rsa, the first and second flanges **100**, **102** may, for example, meet greater resistance when moved by the central wedge **90**. This may limit the extent to which each of the first and second flanges **100**, **102** protrude outwards of the first and second flange bores **42**, **44**, which may negatively affect the clamping between the adapter **20** and the cutting head **120**.
[0102] Each of the first and second flanges **100**, **102** may further include a wedge connector portion **116**. The wedge connector portion **116** is located adjacent to the central wedge **90**. The wedge connector portion **116** connects the forward and rearward facing flange surfaces **106**, **110**. As explained above, the wedge connector portion **116** is configured for sliding movement relative to the central wedge **90** while securing the flange to the central wedge **90**. Such a configuration can be achieved, for example, through a dovetail connection. As shown in the figures, the wedge connector portion **116** may be a female dovetail part. Alternatively, the wedge connector portion **116** may be a male dovetail part, or any configuration which allows relative movement between the central wedge **90** and each of the first and second flanges **100**, **102**.
[0103] In embodiments such as described above, where each of the first and second flanges **100**, **102** includes a wedge connector portion **116** shaped for a dovetail connection, each of the first and second flanges **100**, **102** may comprise a flange dovetail formation **118**. The flange dovetail

formation **118** is located at the wedge connector portion **116**. The flange dovetail formation **118** may be either female or male. Similarly, the central wedge **90** may include two wedge dovetail formations **98**.

[0104] It will be noted that the clamping mechanism **80** may include more flanges than the first and second flanges **100**, **102**. That is to say, the clamping mechanism **80** may include a third flange (not shown), as well as a fourth flange (not shown) and so on. The minimum amount of flanges desired is two, as two flanges can allow for stable fastening. In some embodiments, two flanges may be advantageous in view of facilitated manufacturing, less stringent tolerance requirements and less complex and/or smaller parts.

[0105] Similarly, the central wedge **90** may then include a number of wedge dovetail formations **98** according to the desired number of flanges. Further, the adapter **20** may include a number of flange bores corresponding to the desired number of flanges. For example, in some embodiments, in a clamping mechanism, a central wedge may comprise three or four wedge dovetail formations, and three or four flanges respectively. An adapter according to this embodiment may comprise three or four flange bores respectively. Additionally, there may be a second central wedge (not shown) additional to the central wedge **90**, with the overall number of wedge dovetail formations **98** of all the central wedges being equal to the desired number of flanges.

[0106] Each wedge dovetail formations **98** is in sliding engagement with the flange dovetail formation **118** of any of the first and second flanges **100**, **102** in a direction along the actuating axis A. It will be noted that the direction of the aforementioned sliding engagement is not necessarily parallel to the actuating axis A.

[0107] In some embodiments, the first and second flanges **100**, **102** converge towards one another in the first axial direction D**1** (see FIG. **8**). Differently put, the wedge connector portion **116** of each of the first and second flanges **100**, **102** may slant toward the actuating axis A in the first axial direction D**1**. That is to say, the wedge connector portion **116** of the first flange **100** may slant towards the actuating axis A in the first axial direction D**1** and the wedge connector portion **116** of the second flange **102** may slant towards the actuating axis A in the first axial direction D**1**. Having the first and second flanges **100**, **102**, and more specifically the wedge connector portion **116** of each of the first and second flanges **100**, **102** slanted towards the actuating axis A, facilitates movement of the first and second flanges **100**, **102** in a direction different from the direction in which the central wedge **90** is movable. The direction about which each of the first and second flanges **100**, **102** are movable may be, for example, parallel to the outward radial direction Ro.

[0108] In the embodiments shown in the figures, the central wedge **90** is movable along the actuating axis A, and each of the first and second flanges **100**, **102** is movable in a direction perpendicular thereto. Specifically, the first and second flanges **100**, **102** may be movable in a direction perpendicular to both the actuating axis A and to the central adapter axis Ca.

[0109] In some embodiments, a wedge connector angle wca (shown in FIG. **14***b*) is defined between the actuating axis A and the wedge connector portion **116** of one of the first and second flanges **100**, **102**. Differently put, the wedge connector angle wca is defined as the angle between the actuating axis A and the wedge connector portion **116** from a top view (said top view of the wedge connector portion **116** is best shown in FIGS. **8** and **14***b*). Specifically, the wedge connector portion **116** may define a wedge connector axis Wa, and the wedge connector angle wca may be defined between the actuating axis A and the wedge connector axis Wa. The wedge connector angle wca may be further defined from an axial view of one of the first and second flanges **100**, **102** (see FIG. **14***b*).

[0110] The wedge connector angle wca fulfils the following condition: 5°≤wca≤250. In some embodiments, the wedge connector angle wca fulfils the following condition: 10°≤wca≤20°. The wedge connector angle wca may further fulfil the following condition: 12°≤wca≤18°. The wedge connector angle wca influences, for example, the rate at which the first and second flanges **100**, **102** are displaced upon actuation of the actuator **82**. The above angle ranges may be desirable in consideration of the influences of the wedge connector angle wca.

[0111] In some embodiments, the central wedge **90** may include a first wedge surface **91**, a second wedge surface **92**, a forward wedge surface **93**, a rearward wedge surface **94**, an actuator wedge section **95**, a first flange connector portion **96** and a second flange connector portion **97**. The first wedge surface **91** faces in the first axial direction D**1**. The second wedge surface **92** is located farther in the second axial direction D**2** than the first wedge surface **91**. The second wedge surface **92** further faces in the second axial direction D**2**.

[0112] The forward wedge surface **93** connects the first and second wedge surfaces **91**, **92**. The forward wedge surface **93** faces in the forward adapter direction Fa. The rearward wedge surface **94** is located rearwardly of the forward wedge surface **93** and connects the first and second wedge surfaces **91**, **92**. The rearward wedge surface **94** faces in the rearward adapter direction Ra. The actuator wedge section **95** extends between the first and second wedge surfaces **91**, **92** and engages the actuator **82**. In some embodiments, the actuator wedge section **95** may be recessed into one of the forward and rearward wedge surfaces **93**, **94**. Upon actuation of the actuator **82**, the actuator wedge section **95** which is engaged to the actuator **82** may then transfer the actuation into a movement of the central wedge **90**.

[0113] The first flange connector portion **96** is slidingly connected to the first flange **100**. The first flange connector portion **96** extends between the first and second wedge surfaces **91**, **92**. The first flange connector portion **96** may connect the forward and rearward wedge surfaces **93**, **94**. As mentioned above, the central wedge **90** may include two (or more) wedge dovetail formations **98**. At least one of the aforementioned wedge dovetail formations **98** is located at the first flange connector portion **96**.

[0114] The second flange connector portion **97** is slidingly connected to the second flange **102**. The second flange connector portion **97** extends between the first and second wedge surfaces **91**, **92**. The second flange connector portion **97** may connect the forward and rearward wedge surfaces **93**, **94**. As mentioned above, the central wedge **90** may include two (or more) wedge dovetail formations **98**. At least one of the aforementioned wedge dovetail formations **98** is located at the second flange connector portion **97**.

[0115] In some embodiments, the first and second flange connector portions **96**, **97** slant towards the actuating axis A in the first axial direction D**1**. That is to say, a portion of each of the first and second flange connector portions **96**, **97** that is located adjacent to the first wedge surface **91**, is located closer to the actuating axis A relative to a portion of each of the first and second flange connector portions **96**, **97** that is located adjacent to the second wedge surface **92**. Such a slant may facilitate movement of the first and second flanges **100**, **102**, as mentioned above, in a direction perpendicular to the central adapter axis A. The dovetail wedge formation **98** of each of the first and second flanges **100**, **102** may be located at, respectively, the first and second flange connector portions **96**, **97**.

[0116] In some embodiments a flange connector angle fca is defined between the first and second flange connector portions **96**, **97**. Specifically, the first flange connector portion **96** may define a first flange connector axis Fa**1** and the second flange connector portion **97** may define a second flange connector axis Fa**2**. The flange connector angle fca may be defined between the first and second flange connector axes Fa**1**, Fa**2**. The flange connector angle fca may fulfil the following condition: $10° \leq fca \leq 50°$. The flange connector angle fca may further fulfil the following condition: $20° \leq fca \leq 40°$. Additionally, the flange connector angle fca may fulfil the following condition: $25° \leq fca \leq 35°$. The flange connector angle fca may influence, for example, the rate at which the first and second flanges **100**, **102** are displaced upon actuation of the actuator **82**. The above angle ranges may be desirable.

[0117] In some embodiments, the wedge and flange connector angles wca, fca may fulfil the following condition: $fca = 2 * wca$. The flange connector angle fea is measured between the first and second flange connector portions **96**, **97** while the wedge connector angle wca is measured between

the actuating axis A and the wedge connector portion **116** of one of the first and second flanges **100**, **102**.
[0118] Further reference is now made to FIGS. **15** to **21**, a cutting head **120** is shown. The cutting head **120** defines a central head axis Ch and includes a forward head end section **122**, a rearward head end section **124** and a peripheral head surface **150**. The central head axis Ch defines a forward head direction Fh and a rearward head direction Rh opposite to the forward head direction Fh. The rearward head end section **124** is located farther in the rearward head direction Rh (i.e. rearwardly) of the forward head end section **122**. The rearward head end section **124** includes a head clamping portion **126**. The peripheral head surface **150** connects the forward head end section **122** and the rearward head end section **124**.
[0119] The head clamping portion **126** includes a receptacle wall **132** that defines a head clamping receptacle **130** and an actuator access bore **148**. The head clamping receptacle **130** is recessed into the head clamping portion **126**. The receptacle wall **132** circumferentially bounds the head clamping receptacle **130**, with the head clamping receptacle **130** opening out to the cutting head **120**. The actuator access bore **148** opens out to the cutting head **120**, and more specifically to the peripheral head surface **150**, and to the head clamping receptacle **130**. In some embodiments, the actuator access bore **148** may open out to the head clamping portion **126**, as shown in the figures. Alternatively, the actuator access bore **148** may, for example, open out to the forward head end section **122**. The actuator access bore **148** ensures that a key as discussed above can actuate the actuator **82**. In some embodiments, the head clamping portion **126** may include a head torque transfer mechanism **128**. Such a head torque transfer mechanism **128** may be advantageous for cutting tools, and specifically for rotary cutting tools.
[0120] The receptacle wall **132** includes a forward facing head clamping surface **134** and a rearward facing head abutment surface **136**. The head clamping surface **134** faces in the forward head direction Fh. The head clamping surface **134** extends away from the central head axis Ch and towards the peripheral head surface **150**. Differently put, the head clamping surface **134** is slanted in a direction towards the central holder axis Ch and towards the rearward holder direction Rh. In some embodiments, the head clamping surface **134** is elongated in a direction about the central head axis Ch (i.e. along the central plane Pc or a head plane Pc, as discussed below). The head clamping surface **134** may encircle the central head axis Ch. The head abutment surface **136** faces in the rearward head direction Rh. The head abutment surface **136** may encircle the central head axis Ch.
[0121] In some embodiments, the head clamping surface **134** is cone-shaped (i.e., forms concentric circles, with the diameter of such circles decreasing along the rearward holder direction Rh). Such a shape may be able to be produced easily and cheaply, while also providing good precision for abutment.
[0122] In some embodiments, the head abutment surface **136** delimits the cutting head **120** in the rearward head direction Rh and encircles the head clamping receptacle **130**. Having the head abutment surface **136** located outwardly of the head clamping receptacle **130** may allow for a more stable and/or robust clamping between the adapter **20** and the cutting head **120**.
[0123] In some embodiments, a head clamping angle hca is defined between the head clamping surface **134** and a head plane Ph. The head plane Ph is perpendicular to the central head axis Ch. The head clamping angle hca is inwardly defined between the head clamping surface **134** and the head plane Ph. The head clamping angle hca may fulfil the following condition: $5° \leq hca \leq 22°$. The head clamping angle hca may further fulfil the following condition: $7° \leq hca \leq 17°$. Additionally, the head clamping angle hca may fulfil the following condition: $10° \leq hca \leq 14°$. The values of the head clamping angle hca are related to the values of the rear slanted flange angle rsa. As mentioned above regarding the rear slanted flange angle rsa, the above angle ranges may be desirable due to considerations such as directionality of clamping forces and the degree to which the first and second flanges **100**, **102** may protrude outwards of the first and second flange bores **42**, **44**,

respectively.

[0124] It may be desirable in some embodiments that the head clamping angle hea and the rear slanted flange angle rsa fulfil the following condition: $rsa \leq hca$. The head clamping angle hca and the rear slanted flange angle rsa may further fulfil the following condition: $rsa < hca \leq rsa + 5°$. Having such a condition may, for example, ensure repeatability in the clamping between the first and second flanges **100**, **102** and the head clamping surface **134**.

[0125] In some embodiments, the head abutment surface **136** includes two rear abutting portions **138** and two relief portions **140**. The two relief portions **140** are located farther in the forward holder direction Fh than the two rear abutting portions **138**. Each of the two relief portions **140** are located between the two rear abutting portions **138**. Further in some embodiments, the two relief portions **140** may be located opposite one another about the central holder axis Ch. Dividing the head abutment portion **136** into two abutting portions and two relieved portions may ensure a better distribution of the clamping forces. To ensure that, each of the two relief portions **140** are located directly rearwardly of one of the first and second flanges **100**, **102**. Thus, rather than the first and second flanges **100**, **102** clamping the cutting head **120** directly against the adapter **20**, and focusing the clamping forces in a direction directly rearwardly of them, the clamping forces may be more evenly distributed.

[0126] With further reference to FIGS. **1** and **2**, as well as FIGS. **18**, **19**, **20** and **21**, as mentioned above, the cutting tool **1** includes the above-mentioned adapter **20** and cutting head **120**. The central adapter axis Ca and the central head axis Ch coincide with one another (i.e. overlap and parallel to one another). The forward adapter direction Fa and the forward holder direction Fh align with one another. The rearward adapter direction Ra and the rearward holder direction Rh align with one another. The rear adapter end section **24** may, for example, facilitate clamping of the cutting tool **1** to a turret during machining operations.

[0127] When assembling the cutting tool **1** the cutting head **120** is releasably fastened to the adapter **20** via the clamping mechanism **80**. The adapter clamping receptacle **38** of the adapter **20** is positioned within the head clamping receptacle **120** (i.e. the receptacle wall **132** encircles the adapter clamping receptacle **38**). The adapter clamping portion **34** and the head clamping portion **126** overlap one another. In some embodiments, the adapter torque transfer mechanism **58** engages the head torque transfer mechanism **128**.

[0128] The actuator bore **66** of the adapter **20** overlaps with the actuator access bore **148** of the cutting head **120**. In this manner a key may be inserted through the actuator access bore **148** into engagement with the actuator **82**.

[0129] When the adapter is in the active position (i.e. has been actuated), as best shown in FIG. **20**, the actuator **82** moves the central wedge **90** along the actuating axis A within the hollow portion **40**. The movement of the central wedge **90** causes the first and second flange connector portions **96**, **97** of the central wedge **90** to abut the wedge connector portion **116** of each of the first and second flanges **100**, **102**, respectively, and moves the first and second flanges **100**, **102** in the outward radial direction Ro. In the active position, the first and second flanges **100**, **102** project from the hollow portion **40** and out of the first and second flange bores **42**, **44**, respectively.

[0130] The rear slanted flange surface **112** of each of the first and second flanges **100**, **102** clamps against the head clamping surface **134** of the cutting head **120**, directing clamping forces against the head clamping surface **134** in both the rearward holder direction Rh and in the outward radial direction Ro. In some embodiments, the clamping forces are directed mainly in the rearward holder direction Rh (interchangeable with the rearward adapter direction Ra). The opposite forces arising from the clamping cause the forward and rearward facing flange surfaces **106**, **110** to abut against the forward and rearward bore surfaces **48**, **54**.

[0131] The clamping forces between the head clamping surface **134** and the rear slanted flange surface **112** of each of the first and second flanges **100**, **102** bring the adapter abutment surface **60** into abutment against the head abutment surface **136**. Due to the direction of the clamping forces, such clamping may be very robust. Similarly, due to the rear slanted flange surface **112** of each of the first and second flanges **100**, **102** being elongated as specified, the clamping forces are distributed through at least two points of contact between the rear slanted flange surface **112** of each of the first and second flanges **100**, **102** and the head clamping surface **134**. Such distribution may, for example, lower the forces working against any specific area of the head clamping surface **134** and the first and second flanges **100**, **102**. Additionally, such a distribution may, for example, allow for a more stable clamping, due to at least four different points of contact (at least two points of contact between the cutting head **120** and each of the first and second flanges **100**, **102**) between the adapter **20** and the cutting head **120**.

[0132] In some embodiments, the relief portion **140** of the cutting head **120** is located directly in the rearward head direction Rh relative to the first and second flange bores **42**, **44**. Similarly, the relief portion **140** is located directly rearwardly of the first and second flanges **100**, **102**. Thus, the clamping forces arising from the first and second flanges **100**, **102** clamping against the head clamping surface **134** of the cutting head **120** may be more evenly distributed about the head abutment surface **136**, and more specifically about the rear abutting portion **138** of the head abutment surface **136**. Without the above mentioned relief portion **140** the clamping forces may, for example, be concentrated directly rearwards of the first and second flanges **100**, **102**, which may adversely affect the stability of the clamping between the adapter **20** and the cutting head **120**.

[0133] When the adapter **20** is in the inactive position, as best shown in FIG. **21**, the actuator **20** moves the central wedge **90** axially along the actuating axis A, in a direction opposite to the direction along which the central wedge **90** is moved when the adapter **20** is in the active position. Said movement of the central wedge **90** causes the first and second flange connector portions **96**, **97** of the central wedge **90** to abut the wedge connector portion **116** of each of the first and second flanges **100**, **102**, respectively, and moves the first and second flanges **100**, **102** towards the central adapter axis Ca (i.e. in a direction opposite to the outward radial direction Ro). The first and second flanges **100**, **102** are retracted within (i.e. confined to) the adapter **20**. Differently put, the first and second flanges **100**, **102** do not project out of the first and second flange bores **42**, **44**. The cutting head **120** is not fastened to the adapter **20** in the inactive position and the cutting head **120** is removable from the adapter **20**.

[0134] The peripheral head surface **150** includes at least one of a cutting insert pocket **152** and an integral cutting element (not shown). As specified, the cutting head **120** is used in machining operations, specifically metal machining operations. Thus the cutting head **120** must accommodate a cutting element, for example as discussed above, through the at least one insert pocket **152**.

[0135] In some embodiments, the cutting head **120** may include, at the head clamping portion **126**, a thickened head portion **144**. The thickened head portion **144** cannot participate in the cutting process of machining operations. Differently put, the thickened head portion **144** is devoid of an integral cutting edge or a cutting insert pocket **152**, as discussed above. The peripheral head surface located at the thickened head portion **144** is located farther away from the central head axis Ch relative to the peripheral head surface **152** located at forwardly of the thickened head portion **144**. This is done to accommodate the adapter clamping portion **34** within the head clamping receptacle **130**. The thickened head portion **144** may be necessary in some cutting heads **120** configured for use with bigger adapters **20**. The use of the word “big” herein refers to the dimensions of the adapter **20** relative to the desired dimensions of the cutting head **120**.

[0136] Alternatively, in some embodiments, as shown in FIGS. **22** and **23**, cutting head **120′** is shown. Cutting head **120′** can be considered identical to the cutting head **120**, with the exception that cutting head **120′** is devoid of a thickened head portion **144**. Differently put, the peripheral head surface **150** of the cutting head **120** may coincide with an envelope of a simple geometrical shape, such as, for example as shown in FIGS. **22** and **23**, a cylindrical shape. The above specified thickened head portion **144** may not be necessary in some cases, such as where the relative size of the adapter **20** and the cutting head **120** is similar.

[0137] While several embodiments of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Any combination of the above embodiments is also envisioned and is within the scope of the appended claims. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope of the claims appended hereto.

Claims

1. An adapter (**20**) having a central adapter axis (Ca) defining an outward radial direction (Ro) extending away from the central adapter axis (Ca), opposing forward and rearward adapter directions (Fa, Ra) extending parallel to the central adapter axis (Ca), and a central plane (Pc) orthogonal to the central adapter axis (Ca), the adapter (**20**) comprising: a rear adapter end section (**24**); a forward adapter end section (**30**) located forward of the rear adapter end section (**24**) and a peripheral adapter surface (**70**) connecting the forward adapter end section (**30**) and the rear adapter end section (**24**); the forward adapter end section (**30**) including an adapter clamping portion (**34**), the adapter clamping portion (**34**) comprising an adapter clamping receptacle (**38**), the adapter clamping receptacle (**38**) defining: a first flange bore (**42**) opening out to the peripheral adapter surface (**70**); a second flange bore (**44**) opening out to the peripheral adapter surface (**70**); and an actuator bore (**66**) extending into the adapter clamping receptacle (**38**); and a clamping mechanism (**80**) comprising: an actuator (**82**) defining an actuating axis (A) having opposite first and second axial directions (D**1**, D**2**) extending parallel to the actuating axis (A), the actuator (**82**) being aligned with the actuator bore (**66**) and being at least partially located in the adapter clamping receptacle (**38**); a central wedge (**90**) connected to the actuator (**82**); a first flange (**100**) slidingly connected to the central wedge (**90**) and configured to project through the first flange bore (**42**); and a second flange (**102**) opposite the first flange (**100**) about the actuating axis (A), the second flange (**102**) slidingly connected to the central wedge (**90**) and configured to project through the second flange bore (**44**); each of the first and second flanges (**100**, **102**) comprising: a forward facing flange surface (**106**); and a rearward facing flange surface (**110**), the rearward facing flange surface (**110**) comprising a rear slanted flange portion (**112**), the rear slanted flange portion (**112**) extending forwardly in the outward radial direction (Ro), the rear slanted flange portion (**112**) elongated in a direction along the central plane (Pc).

2. The adapter (**20**) according to claim 1, wherein the rear slanted flange portion (**112**) is cone shaped.

3. The adapter (**20**) according to claim 1, wherein each of the first and second flanges (**100**, **102**) comprise a wedge connector portion (**116**) located adjacent to the central wedge (**90**), the wedge connector portion (**116**) connecting the forward and rearward facing flange surfaces (**106**, **110**), the wedge connector portion (**116**) of the first flange (**100**) slants towards the actuating axis (A) in the first axial direction (D**1**) and the wedge connector portion (**116**) of the second flange (**102**) slants towards the actuating axis (A) in the first axial direction (D**1**).

4. The adapter (**20**) according to claim 3, wherein a wedge connector angle wca, defined between the actuating axis (A) and the wedge connector portion (**116**) of one of the first and second flanges (**100**, **102**), fulfils the following condition: $5° \leq wca \leq 25°$.

5. The adapter (**20**) according to claim 3, wherein each of the first and second flanges (**100**, **102**) comprise, at the wedge connector portion (**116**), a flange dovetail formation (**118**), the central wedge (**90**) comprises two wedge dovetail formations (**98**) engaging with the flange dovetail formations (**118**); each of the flange dovetail formations (**118**) in sliding engagement with one of the two wedge dovetail formations (**98**) in a direction along the actuating axis (A).

6. The adapter (**20**) according to claim 1, wherein the actuator (**82**) has an active position in which the first and second flanges (**100**, **102**) project radially from the first and second flange bores (**42**, **44**), respectively, wherein the actuator (**82**) has an inactive position in which the first and second flanges (**100**, **102**) do not project radially from the first and second flange bores (**42**, **44**), respectively.

7. The adapter (**20**) according to claim 6, wherein each of the first and second flanges (**100**, **102**) further comprises an outer flange surface (**114**) facing away from the actuating axis (A), wherein in the inactive position, the outer flange surface (**114**) of each of the first and second flanges (**100**, **102**) is level with the peripheral adapter surface (**70**) adjacent to the respective one of the first and second flange bores (**42**, **44**).

8. The adapter (**20**) according to claim 1, wherein a rear slanted flange angle rsa, defined between the central plane (Pc) and the rear slanted flange portion (**112**), fulfils the following condition: 3°≤rsa≤20°.

9. The adapter (**20**) according to claim 1, wherein the clamping mechanism (**80**) further comprises an actuator captive mechanism (**84**) securing the actuator (**82**) in the adapter (**20**).

10. The adapter (**20**) according to claim 1, wherein the rearward facing flange surface (**110**) is planar.

11. The adapter (**20**) according to claim 1, wherein the first and second flanges (**100**, **102**) are identical.

12. The adapter (**20**) according to claim 1, wherein the central wedge (**90**) comprises: a first wedge surface (**91**); a second wedge surface (**92**) located farther in the second axial direction (D**2**) than the first wedge surface (**91**); a forward wedge surface (**93**) connecting the first and second wedge surfaces (**91**, **92**); a rearward wedge surface (**94**) located rearwardly of the forward wedge surface (**93**) and connecting the first and second wedge surfaces (**91**, **92**); an actuator wedge section (**95**) extending between the first and second wedge surfaces (**91**, **92**) and engaged to the actuator (**82**); a first flange connector portion (**96**) slidingly connected to the first flange (**100**), the first flange connector portion (**96**) extending between the first and second wedge surfaces (**91**, **92**); and a second flange connector portion (**97**) slidingly connected to the second flange (**102**), the second flange connector portion (**97**) extending between the first and second wedge surfaces (**91**, **92**).

13. The adapter (**20**) according to claim 12, wherein the first flange connector portion (**96**) slants towards the actuating axis (A) in the first axial direction (D**1**) and the second flange connector portion (**97**) slants towards the actuating axis (A) in the first axial direction (D**1**).

14. The adapter (**20**) according to claim 13, wherein a flange connector angle fca, defined between the first and second flange connector portions (**96**, **97**), fulfils the following condition: 10°≤fca≤50°.

15. The adapter (**20**) according to claim 14, wherein the wedge and flange connector angles wca, fca fulfil the following condition: fca=2*wca.

16. The adapter (**20**) according to claim 14, wherein each of the first and second flanges (**100**, **102**) comprises a wedge dovetail formation (**98**).

17. A cutting tool (**1**) comprising an adapter (**20**) according to claim 1; and a cutting head (**120**) attached to the adapter (**20**), the cutting head (**120**) defining a central head axis (Ch) coinciding with the central adapter axis (Ca) of the adapter (**20**) and having forward and rearward head directions (Fh, Rh) extending parallel to the central head axis (Ch).

18. The cutting tool (**1**) according to claim 17, wherein the cutting head (**120**) comprises: a forward head end section (**122**); a rearward head end section (**124**) located rearward of the forward head end section (**122**) and a peripheral head surface (**150**) connecting the forward head end section (**122**) and the rearward head end section (**124**); the reward head end section (**124**) comprising a head clamping portion (**126**), the head clamping portion (**126**) comprising: a receptacle wall (**132**) comprising a forward facing head clamping surface (**134**) and a rearward facing head abutment surface (**136**), the receptacle wall (**132**) defining: a head clamping receptacle (**130**) recessed into the head clamping portion (**126**); and an actuator access bore (**148**) opening out to the peripheral head surface (**150**) and to the head clamping receptacle (**130**); the peripheral head surface (**150**) comprising an integral cutting element or defining a cutting insert pocket (**152**) configured to receive a cutting element; wherein in an active position of the actuator (**82**) of the adapter (**20**), the rear slanted flange portion (**112**) abuts the forward facing head clamping surface (**134**).

19. A cutting head (**120**) defining a central head axis (Ch) defining forward and rearward head directions (Fh, Rh), the cutting head (**120**) comprising: a forward head end section (**122**); a rearward head end section (**124**) located rearward of the forward head end section (**122**) and a peripheral head surface (**150**) connecting the forward head end section (**122**) and the rearward head end section; the rearward head end section (**124**) including a head clamping portion (**126**), the head clamping portion (**126**) comprising a receptacle wall (**132**), the receptacle wall (**132**) including a forward facing head clamping surface (**134**) and a rearward facing head abutment surface (**136**), the head clamping surface (**134**) extending in a direction away from the central head axis (Ch) and in the forward head direction (Fh), the receptacle wall (**132**) defining: a head clamping receptacle (**130**) recessed into the head clamping portion (**126**); and an actuator access bore (**148**) opening out to the peripheral head surface (**150**)) and to the head clamping receptacle (**130**); the peripheral head surface (**150**) including an integral cutting element or defining a cutting insert pocket (**152**) configured to receive a cutting element.

20. The cutting head (**120**) according to claim 19, wherein the head clamping surface (**134**) is elongated in a direction about the central head axis (Ch).

21. The cutting head (**120**) according to claim 20, wherein the head clamping surface (**134**) encircles the central head axis (Ch).

22. The cutting head (**120**) according to claim 19, wherein the head clamping surface (**134**) is cone shaped.

23. The cutting head (**120**) according to claim 19, wherein a head clamping angle hca, defined between the head clamping surface (**134**) and a head plane (Ph), the head plane (Ph) being perpendicular to the central head axis (Ch), fulfils the following condition: $5° \leq hca \leq 22°$.

24. The cutting head (**120**) according to claim 19, wherein the head abutment surface (**136**) comprises two rear abutting portions (**138**) and two relief portions (**140**) located forwardly of the two rear abutting portions (**138**), each of the two relief portions (**140**) being located between the two rear abutting portions (**138**).

25. A cutting tool (**1**) comprising: an adapter (**20**); and the cutting head (**120**) according to claim 19.

26. The cutting tool (**1**) according to claim 25, wherein the adapter (**20**) has a central adapter axis (Ca) defining an outward radial direction (Ro) extending away from the central adapter axis (Ca), opposing forward and rearward adapter directions (Fa, Ra) extending parallel to the central adapter axis (Ca), and a central plane (Pc) orthogonal to the central adapter axis (Ca), the adapter (**20**) comprising: a rear adapter end section (**24**); a forward adapter end section (**30**) located forward of the rear adapter end section (**24**) and a peripheral adapter surface (**70**) connecting the forward adapter end section (**30**) and the rear adapter end section (**24**); the forward adapter end section (**30**) including an adapter clamping portion (**34**), the adapter clamping portion (**34**) comprising an adapter clamping receptacle (**38**), the adapter clamping receptacle (**38**) defining: a first flange bore (**42**) opening out to the peripheral adapter surface (**70**); a second flange bore (**44**) opening out to the peripheral adapter surface (**70**); and an actuator bore (**66**) extending into the adapter clamping receptacle (**38**); and a clamping mechanism (**80**) comprising: an actuator (**82**) defining an actuating axis (A) having opposite first and second axial directions (D**1**, D**2**) extending parallel to the actuating axis (A), the actuator (**82**) being aligned with the actuator bore (**66**) and being at least partially located in the adapter clamping receptacle (**38**); a central wedge (**90**) connected to the actuator (**82**); a first flange (**100**) slidingly connected to the central wedge (**90**) and configured to project through the first flange bore (**42**); and a second flange (**102**) opposite the first flange (**100**) about the actuating axis (A), the second flange (**102**) slidingly connected to the central wedge (**90**) and configured to project through the second flange bore (**44**); each of the first and second flanges

(**100**, **102**) comprising: a forward facing flange surface (**106**); and a rearward facing flange surface (**110**), the rearward facing flange surface (**110**) comprising a rear slanted flange portion (**112**), the rear slanted flange portion (**112**) extending forwardly in the outward radial direction (Ro), the rear slanted flange portion (**112**) elongated in a direction along the central plane (Pc).